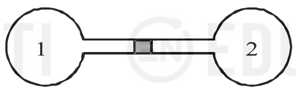
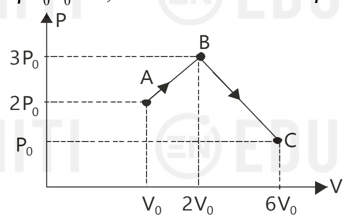
Subjective Type

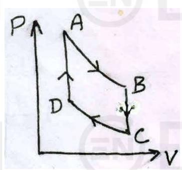
1. Two spherical flasks having volume $V_0 = 1.0$ L each containing air are connected by a tube of diameter d = 6 mm and length l = 1m. A small droplet of mercury contained in the tube is at its middle at 0°C. By what distance do the mercury droplet move if the flask 1 is heated by 2°C while flask 2 is cooled by 2°C. Ignore any expansion of flask wall.



2. One mole of monatomic ideal gas undergoes a process ABC as shown in figure. If the maximum temperature of the gas during the process ABC is $\beta P_0V_0/R$, then the value of β is



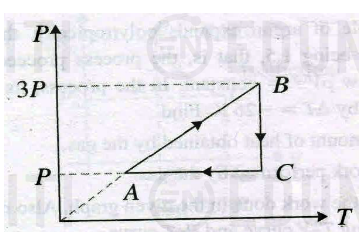
3. On mole of a monatomic ideal gas is taken through the cycle shown in figure.



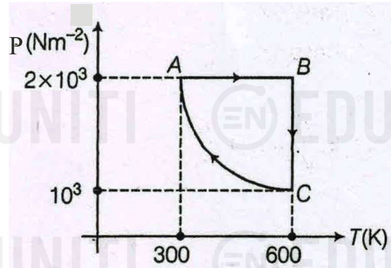
Processes A → B and C → D are adiabatic and Processes B → C and D → A are isochoric.
Given that T_A = 1000 K, P_B = (2/3) P_A and P_C = (1/3) P_A.
Calculate the following quantities:
(a) The work done by the gas in the process A → B
(b) The heat lost by the gas in the process B → C
(c) The temperature T_D.
(Take $(2/3)^{2/5} = 0.85$)

4. A massles piston divides a closed, thermally insulated cylinder into two equal parts. One part contains 28g of nitrogen. At this temperature, one-third of molecules are dissociated into atoms and the other part is evacuated. The piston is released and the gas fills the whole volume of the cylinder at temperature T_0. Then, the piston is slowly displaced back to its initial position. Calculate the increase in internal energy of the gas. Neglect further dissociation of molecules during, the motion of the piston.

5. A fixed mass of oxygen gas performs a cyclic process *ABCA* as shown. Find the efficiency of the process.

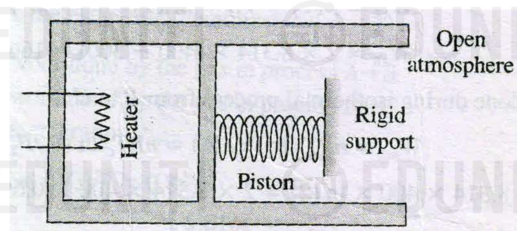


6. Two moles of a monatomic ideal gas is taken through a cyclic process as shown.

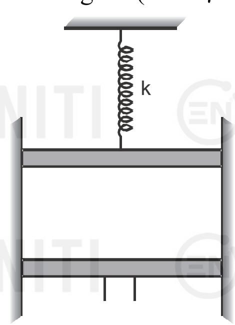


Process *CA* follows *PT* = constant. Find the cycle efficiency

7. An ideal monatomic gas is confined in a cylinder by a massless and frictionless spring loaded piston of cross sectional area 8×10^{-3} m^2. Initially, the gas is at 300 K and occupies a volume of 2.4×10^{-3} m^3 and the spring is in its relaxed state. Now the gas is heated by a small electric heater until the piston moves out slowly by 0.1 m. Calculate the final temperature of the gas, the change in internal energy of the gas, the work done by the gas and the heat supplied by the heater. The cylinder and the piston are thermally insulated. The force constant of the spring is 8000 N/m and atmospheric pressure is $P_0 = 1 \times 10^5$ N/m^2.



8. An ideal gas at NTP is enclosed in a adiabatic vertical cylinder having area of cross section A = 27 cm^2, between two light movable adiabatic pistons as shown in the figure. Spring with force constant k = 3700 N/m is in a relaxed state initially. Now the lower piston is moved upwards by h/2 where h is the initial length of gas column. It is observed that the upper piston moves up by a distance h/16. Find h and the final temperature of the gas. (Take $\gamma = 1.5$)

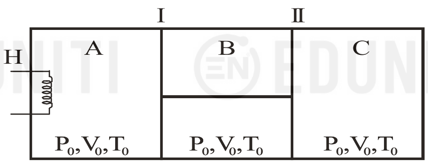


9. The figure shows an insulated cylinder divided into three parts, A, B and C. Pistons I and II are connected by a rigid rod and can move without friction inside the cylinder. Piston I is perfectly conducting while piston II is perfectly insulating. The initial state of the gas $(\gamma = 1.5)$ present in each compartment A, B and C is as shown. Now, compartment A is slowly given heat through a heater H such that the final volume of C becomes $4V_0/9$. Assuming the gas to be ideal, find

(a) final pressures and final temperatures in each compartment A,B and C

(b) heat supplied by the heater

(c) work done by gas in A and B and heat flowing across piston I



Paragraph Type

Passage for Question Nons. 10 to 12

An ideal diatomic gas is expanded so that the amount of heat transferred to the gas is equal to the decrease in its internal energy.

10. The molar specific heat of the gas in this process is given by C whose value is

(a) $-5R/2$ (b) $-3R/2$
(c) $2R$ (d) $5R/2$

11. The process can be represented by the equation TV^n = constant, where the value of n is

(a) 7/5 (b) 1/5
(c) 3/2 (d) 3/5

12. If in the above process, the initial temperature of the gas is T_0 and the final volume is 32 times the intial volume, the work done (in joules) by one mole of gas during the process will be

(a) $\frac{RT_0}{2}$ (b) $\frac{5RT_0}{2}$
(c) $2RT_0$ (d) RT_0

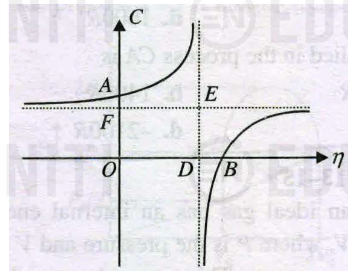
Passage for Question Nons. 13 to 15

A process in which work performed by an ideal gas is proportional to the corresponding change in its internal energy is described as a polytropic process. If we represent work done by a polytropic process by ΔW and increase in internal energy as ΔU, then $\Delta W \propto \Delta U$

For this process, it can be shown that the relation between pressure and volume is given by the equation

$$PV^{\alpha} = \text{constant}$$

In polytropic process, the variation of molar specific heat C with α for a monatomic gas is plotted as shown in the graph.



13 In the graph, the y-coordinate of point A is

(a) $3R/2$ (b) $5R/2$
(c) $7R/2$ (d) $4R$

14. In the graph, the x-coordinate of point B is

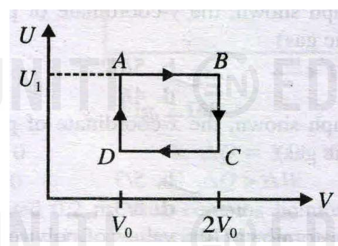
(a) 7/5 (b) 5/3
(c) 2/3 (d) 8/3

15. For a monatomic gas, the values of polytropic constant α for which molar specific heat is negative is

(a) $1 < \alpha < 2/3$ (b) $1 < \alpha < 8/3$
(c) $1 < \alpha < 5/3$ (d) $2/3 < \alpha < 8/3$

Passage for Question Nons. 16 to 18

One mole of an ideal gas has an internal energy given by $U = U_0 + 2PV$, where P is the pressure and V is the volume of the gas. U_0 is a constant. This gas undergoes the quasi-static cyclic process $ABCD$ as shown in the $U - V$ diagram.



16. The molar heat capacity of the gas at constant pressure is

(a) $2R$ (b) $3R$
(c) $\frac{5}{2}R$ (d) $4R$

17. The work done by the ideal gas in the process AB is

(a) zero (b) $U_1 - U_0$
(c) $\frac{U_1 - U_0}{2}$ (d) $\left(\frac{U_1 - U_0}{2}\right)\ln 2$

18. The gas must be

(a) monatomic
(b) diatomic
(c) a mixture of mono and diatomic gases
(d) a mixture of di - and tri-atomic gases

Passage for Question Nons. 19 to 20

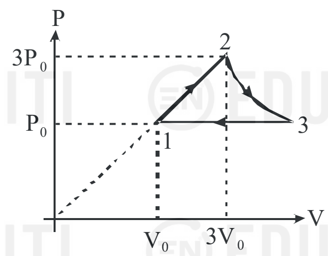
A fluid system undergoes a process following $p = \frac{5}{V} + 1.5$, where p is in bar and V is in m^3. During the process, volume changes from 0.15 m^3 to 0.05m^3 and the system rejects 45 kJ of heat.

19. Work done during the process is (Take, $\log_e 3 = 1.098$)
(a) -564 kJ (b) 564 kJ
(c) 483 kJ (d) -483 kJ

20. Change in internal energy during the process is
(a) -45 kJ (b) 45 kJ
(c) 519 kJ (d) -519 kJ

Passage for Question Nons. 21 to 23

One mole of an ideal monoatomic gas undergoes a cyclic process as shown in figure. Temperature at point 1 is 300 K and process $2 \rightarrow 3$ is isothermal.



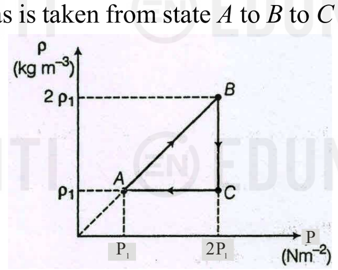
21. Net work done by gas in complete cycle is
(a) $(9\ell n3 + 12)P_0V_0$ (b) $(9\ell n3 + 4)P_0V_0$
(c) $(9\ell n3 - 4)P_0V_0$ (d) $(9\ell n3 - 8)P_0V_0$

22. Heat capacity of process $1 \rightarrow 2$ is
(a) $\frac{R}{2}$ (b) $\frac{3R}{2}$
(c) $\frac{5R}{2}$ (d) 2R

23. The efficiency of cycle is
(a) $\left(\frac{9\ell n3 + 4}{9\ell n3 + 12}\right)$ (b) $\left(\frac{9\ell n3 - 4}{9\ell n3 + 12}\right)$
(c) $\left(\frac{9\ell n3 - 4}{9\ell n3 + 16}\right)$ (d) $\left(\frac{9\ell n3 + 12}{9\ell n3 + 16}\right)$

Passage for Question Nons. 24 to 25

Consider a thermodynamic cycle as shown in which a monatomic ideal gas is taken from state *A* to *B* to *C* to *A*.

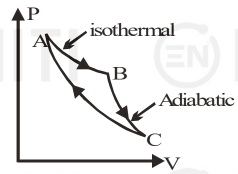


24. Ratios of work done in the processes, *AB*, *BC*, and *CA* $(W_{AB} : W_{BC} : W_{CA})$ are
(a) ln 2 : 1 : 1 (b) $-$ln 2 : 1 : 0
(c) $-$ln 2 : 1 : 0 (d) $-$ln 2 : 1 : 1

25. Ratio of heat rejected by gas to the total heat supplied is
(a) $\frac{2}{5}(1 - \ln 2)$ $\frac{5}{2}(1 - \ln 2)$
(c) $\frac{2}{5}\ln 2$ (d) $\left(\frac{3 + \ln 4}{5}\right)$

Passage for Question Nons. 26 to 27

A cyclic process for an ideal gas is shown in figure. Given $\Delta W_{AB} = +700\,\text{J}, \Delta W_{BC} = +400\,\text{J}, \Delta Q_{CA} = -100\,\text{J}$.

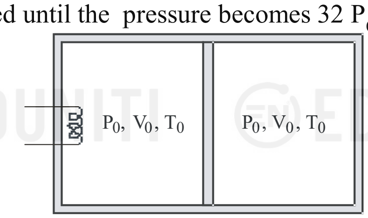


26. The work done in process CA is
(a) -500 J (b) 500 J
(c) 400 J (d) -400 J

27. The efficiency of the cycle is
(a) 100% (b) 83.44%
(c) 85.71% (d) 81.11%

Passage for Question Nons. 28 to 30

One mole of a monoatomic ideal gas occupies two chambers of a cylinder partitioned by means of a movable and frictionless piston. The walls of the cylinder as well as the piston are thermal insulators. Initially equal amounts of gas fill both the chambers at (P_0, V_0, T_0). The left chamber is slowly heated by an electric heater. The gas in the left chamber expands, pushing the piston to the right. The gas on the right chamber is compressed until the pressure becomes 32 P_0.



28. The final volume of left chamber is
(a) $\frac{V_0}{8}$ (b) $\frac{15}{8}V_0$
(c) $\frac{7}{8}V_0$ (d) $\frac{9}{8}V_0$

29. The work done on the gas in the right chamber is
(a) $\frac{9}{2}P_0V_0$ (b) $-\frac{9}{2}P_0V_0$
(c) $\frac{13}{2}P_0V_0$ (d) $\frac{17}{2}P_0V_0$

30. The change in internal energy of the gas in the left chamber is
(a) $\frac{131}{2}P_0V_0$ (b) $\frac{177}{2}P_0V_0$
(c) $59\,P_0V_0$ (d) $93\,P_0V_0$

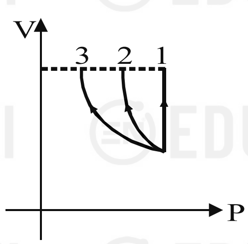
More than one Correct

31. C_p is always greater than C_V due to the fact that
(a) no work is done on heating the gas at constant volume.
(b) when a gas absorbs heat at constant pressure, its volume must change so as to do some external work.
(c) for the same rise of temperature, the internal energy of a gas changes by equal amount at constant volume and at constant pressure.
(d) for the same rise of temperature, the internal energy of a gas changes by a smaller amount at constant volume than at constant pressure.

32 An ideal gas can be expanded from an initial state to a certain volume through two different processes
(i) PV^2 = constant and
(ii) $P = KV^2$ where K is a positive constant.
Then,
(a) final temperature in (i) will be greater than in (ii)
(b) final temperature in (ii) will be greater than in (i)
(c) total heat given to the gas in (i) case is greater than in (ii)
(d) total heat given to the gas in (ii) case is greater than in (i)

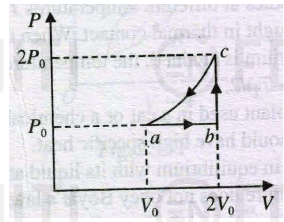
33 n moles of an ideal triatomic linear gas undergoes a process in which the temperature changes with volume as $T = k_1V^2$ where k_1 is a constant. Choose the correct alternatives
(a) At normal temperature $C_v = \frac{5}{2}R$
(b) At any temperature $C_p - C_v = R$
(c) At normal temperature molar heat capacity C = 3R
(d) At any temperature molar heat capacity C=3R

34 The figure shows V-P curve for three processes. Choose the correct statement(s)
(a) Work done is maximum in process 1.



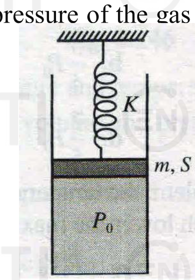
(b) Temperature must increase in process 2 and 3.
(c) Heat must be supplied in process 1.
(d) If final volume of gas in process 1, 2 and 3 are same, then temperature must be same.

35 One mole of an ideal monatomic gas having initial temperature T_0 at a, is made to go through the cycle abca shown in fig. If U denotes the internal energy, then choose the correct alternative/s



(a) $U_c > U_b > U_a$
(b) $U_c - U_b = 3RT_0$
(c) $U_c - U_a = \frac{9RT_0}{2}$
(d) $U_b - U_a = \frac{3RT_0}{2}$

36 In the arrangement shown in fig., gas is thermally insulated. An ideal gas is filled in the cylinder having pressure P_0(> atmospheric pressure P_a). The spring of force constant K is initially unstretched. The piston of mass m and area A is frictionless. In equilibrium, the piston rises up by distance x_0, its speed is v and the pressure of the gas is P. Then,



(a) $P = P_a + \frac{Kx_0}{A} + \frac{mg}{A}$
(b) work done by the gas from initial to equilbrium position is $\frac{1}{2}Kx_0^2 + mgx_0 + \frac{1}{2}mv^2$
(c) decrease in internal energy of the gas from initial to equilbrium position is $P_aAx_0 + \frac{1}{2}Kx_0^2 + mgx_0 + \frac{1}{2}mv^2$
(d) all of the above